The Closure Loop Gas Equation of State: Structural Completeness and the Relic-Abundance Frontier

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April, 2026

1. Introduction: The Ontological Inversion and the Limits of Closure

The trajectory of contemporary theoretical physics has encountered a definitive structural impasse, formally identified within advanced theoretical taxonomies as the "Crisis of Distinction".[1] This profound schism represents the persistent, century-long failure of modern science to reconcile the smooth, deterministic, and continuous geometric manifolds that characterize general relativity with the discrete, probabilistic, and jump-like excitations inherent to quantum mechanics.[1] For decades, the scientific community has been consumed by attempts to force a reconciliation between these two dominant pillars, typically by searching for a hypothetical graviton to quantize gravity, or conversely, by attempting to smooth quantum wavefunctions into a geometric continuum.[1] Rigorous theoretical analysis suggests that this failure is not merely a mathematical deficiency or a lack of experimental precision, but a profound ontological flaw rooted in a substance-based paradigm.[1] Standard cosmological models rely implicitly upon a "Linear Stack" ontology—a hierarchical worldview that privileges static nouns, persistent particles, and immutable fields over active verbs, which encompass fluid transformations and recursive constraint propagation.[1]

The framework of closure ontology and the Nexus Recursive Harmonic Architecture resolves this impasse by executing a radical ontological inversion.[1] It posits that the physical universe is not a spatial container holding discrete objects, but rather a fluid mathematical medium composed entirely of pure recursive operations.[1] Within this process-first ontology, physical laws are not fixed mandates but emergent configurations, and matter is reclassified as a curvature trace left by the processing of information on a high-dimensional lattice.[1]

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The reconstruction of reality as an operational manifold begins with three co-present foundational primitives known as the Δ, Γ, and I Triplex.[1] These primitives are not merely descriptive labels but the strictly necessary mathematical conditions for the emergence of any stable physical structure:

- Δ **(Difference or Gap)**: The absolute condition of distinguishability and contrast.[1] Without differentiation, nothing exists to be measured or analyzed.
- Γ **(Touch or Interface)**: The possibility of relation and the initiating trigger for operational closure loops.[1]
- I **(Conservation or Invariant)**: The condition of persistence and identity.[1] Without a mechanism for invariance, no structural pattern can survive entropic decay.

In this paradigm, macroscopic entities identified as solid objects do not fundamentally exist as discrete, intrinsic units. Instead, they are secondary labels applied to highly stable, recurring gap-patterns—a concept formalized as the "Carbon Glyph" or the hashed output of the universal computation.[1] Physical motion is reinterpreted not as the translation of an object through an empty void, but as the propagation of operational gaps traversing an informational lattice.[1] A single, completed closure loop serves as an operational audit log of constraint satisfaction.[1] The boundary event triggers a contact processed through the substrate's kernel, writing a stored closure record.[1] This record carries an accumulated energetic cost, which constitutes the fundamental source of gravity.[1]

While the foundational derivations demonstrate a highly sophisticated macro-geometric mapping, it is critical to state the limits of the theory's current status accurately. The theoretical structure is closed at the macroscopic level, but the microscopic population origin of the loop gas remains an open frontier.[1] The theory is structurally closed enough that the remaining frontier is essentially one abundance inequality, specifically requiring that the relic-abundance inversion of the operational loops be checked numerically or semi-analytically.[1] The following analysis meticulously maps the established, mathematically closed macro-structure before dissecting the exact parameters of the open relic-abundance frontier.

2. Macro-Geometric Closure and Admissible Geometry

The reconstruction of general relativity necessitates a precise bridge from the microscopic loop activity to the macroscopic stress-energy tensor $T_{\mu\nu}$.[1] The source of gravity is persistent closure trace—the coarse-grained density of completed operational loops within a mesoscale volume V_{cg}.[1] Because geometry cannot couple universally to a scalar density alone, the microscopic source map must project into a symmetric, conserved rank-2 tensor.[1]

To guarantee that the large-scale law governing geometry is universally stable and strictly consistent with the micro-scale conservation of closure activity, the framework imposes four rigorous constraints on the geometric tensor $G_{\mu\nu}$ [1]:

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1. **Locality and Differentiability**: The governing law must manifest as a partial differential equation in the metric $g_{\mu\nu}$ and its derivatives.[1]
2. **Covariance**: The law must take the form of a tensor equation, ensuring there is no preferred frame of reference and that the operational bookkeeping is universal across all observers.[1]
3. **Second-Order Nature**: The equation must contain at most second derivatives of the metric. This is vital for the stability of the physical system, directly preventing the emergence of "ghost" instabilities that plague higher-derivative gravity theories.[1]
4. **Divergence-Free Identity**: The tensor must satisfy the Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ identically to couple consistently to a conserved source.[1]

According to Lovelock's theorem, originally formulated in 1971, in exactly four spacetime dimensions, the only symmetric, divergence-free, second-order tensor that can be constructed from the metric and its first two derivatives is a linear combination of the Einstein tensor and the metric itself.[1] By absorbing the integration constants into the gravitational coupling constant κ and setting the zero-point term to the cosmological constant Λ, the framework uniquely and mandatorily arrives at the Einstein field equations.[1]

The Einstein-class macro geometry is therefore not a theoretical choice or an imposed assumption; it is mathematically forced by the Lovelock constraints.[1] It is the only admissible closure class for a gap-first ontology operating at a macroscopic scale.[1]

The Dual-Null Source Split and Minimal Dual Action

Specifying the exact mapping to the stress-energy tensor requires defining the internal structure of a single closure loop via its action $S[\Psi]$.[1] This action must be constructed exclusively from the reparameterization-invariant geometric invariants of a closed loop embedded in four-dimensional spacetime.[1] The derivation identifies a strictly minimal dual action composed of two essential terms [1]:

$$S[\Psi] = S_{NG} + S_{bulk}$$

1. **The Nambu-Goto Term (S_{NG})**: This term represents the worldsheet area and the inherent string tension (σ_T) of the loop.[1] It accounts for the kinetic and thermal sectors of the macroscopic loop gas. In the non-relativistic limit, it yields the standard dust equation of state ($w = 0$), while in the ultra-relativistic (Hagedorn) limit, it produces the radiation equation of state ($w = 1/3$).[1]
2. **The Bulk Term (S_{bulk})**: To recover the vacuum sector, the Nambu-Goto action alone is insufficient, as the Casimir energy of a closed string produces a vacuum pressure that fails the Lorentz-invariant symmetry requirement.[1] Consequently, a bulk term is introduced, representing the three-volume

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enclosed by the loop, governed by a bulk tension constant Λ_0.[1]

The dual-null source split is formally closed by exact metric variation of this minimal action.[1] The variation of the Nambu-Goto term yields the perfect-fluid form for physical matter, while the variation of the bulk term embedded in four dimensions yields a stress-energy tensor directly proportional to the metric, perfectly deriving the cosmological constant behavior.[1] Furthermore, minimal EFT action uniqueness is guaranteed through Gauss-Bonnet topological considerations and derivative expansions, which suppress higher-order extrinsic curvature (rigidity) terms at low energies.[1]

The Jüttner Matter Equation of State and Sector Interpolation

The thermodynamic state of the closure loop gas dictates the specific relationship between pressure and density across different cosmological epochs.[1] The equation of state (EOS) parameter $w(z)$, where $z = m_\Psi c^2/(k_B T_s)$, successfully and smoothly interpolates between distinct physical matter sectors by treating each completed closure loop as a quasi-particle evaluated through the grand canonical partition function.[1]

Sector	**Thermal Condition**	**EOS Parameter (w)**	**Physical Manifestation**
Cold Matter	$z \gg 1$	$w \approx 0$	Non-relativistic Maxwell-Boltzmann dust
Warm Matter	$z \sim 1$	$0 < w < 1/3$	Exact Bessel interpolation (Synge 1957)
Radiation	$z \ll 1$	$w = 1/3$	Ultra-relativistic Bose/Fermi limit

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Vacuum	Bulk-dominated	$w = -1$	Ground-state volumetric bulk tension

The exact Maxwell-Jüttner matter EOS guarantees structural consistency for the thermal loop gas across the cold, warm, and highly relativistic sectors.[1] The Jüttner distribution accurately accounts for the speeds of particles in a hypothetical gas where the effects of special relativity are non-negligible, ensuring the smooth interpolation of the w parameter.[1]

Crucially, the vacuum equation of state ($w = -1$) and the effective constancy of the cosmological constant (Λ_{eff}) are mathematically closed via two independent proofs.[1] Route B utilizes a symmetry-based argument, requiring the vacuum closure density to be Lorentz invariant, which naturally forces $P = -\rho$.[1] Route A provides a dynamical bulk proof, evaluating the zero-temperature limit of the loop gas pressure.[1] As the temperature approaches zero, kinetic and Casimir pressures vanish, leaving only the strictly constant bulk tension.[1] The convergence of these two independent routes structurally closes the vacuum sector, demonstrating that dark energy is not an external postulation but an emergent property of the unresolvable internal volume of the substrate's computational loops.[1]

The "stiff" matter sector ($w = 1$), representing a maximal causal equation of state, falls entirely outside the current valid regime of the dual action and has been formally and honestly dropped from the operational mapping.[1]

3. Phenomenological Recovery of General Relativity

By establishing the unique closure law through Lovelock constraints and the minimal dual action, the Nexus architecture allows for the direct recovery of the classical phenomena of general relativity, reinterpreted fundamentally as standard limits of the operational substrate.[1]

In the static, weak-field, and slow-motion limit, the metric perturbations reduce the Einstein field equations to the Poisson equation, naturally recovering the Newtonian inverse-square law.[1] In this framework, the inverse-square law is not an intrinsic force propagating through an empty void; it is the natural geometric relaxation of the computational substrate outside an active source of closure processing.[1]

Similarly, the Equivalence Principle is cleanly recovered by analyzing the test-particle action.[1] Because the mass parameter identically cancels from the geodesic equations, inertial and gravitational mass are demonstrated to be strictly identical.[1] The closure ontology interprets this phenomenon simply as "the same

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closure burden viewed from two perspectives"—the accumulation of operational trace inherently alters the trajectory of adjacent gap propagations.[1]

The bending of light is explained as null rays propagating along the rewritten boundary of the substrate.[1] The substrate updates affect both the temporal and spatial paths of the light, exactly doubling the deflection angle predicted by Newtonian corpuscular gravity, which is the defining signature of metric curvature.[1] Gravitational waves, confirmed by observations such as GW170817 to travel at the speed of light, are reinterpreted as propagating boundary-geometry updates.[1] Because substrate updates share the identical causal structure as any other fundamental closure propagation, they naturally propagate at c.[1]

4. The Internal Thermodynamics: Hagedorn Regime and the Density of States

Before addressing the cosmological occupancy of the universe, it is imperative to rigorously define the internal thermodynamic limits of the individual operational loops. This is accomplished within the C.4 mode spectrum and Hagedorn regime derivations, which govern the internal excitation states of the closure loop gas as its temperature approaches a theoretical asymptotic maximum.[1]

Small transverse deformations of a bulk-stabilized Nambu-Goto loop, denoted as $R(\sigma) = R_0 + \xi_n \cos(n\sigma)$, exhibit squared frequencies defined by the relation $\omega_n^2 = n^2/R_0^2 + m_{bulk}^2$.[1] The bulk mass gap is intrinsically tied to the substrate constants via $m_{bulk}^2 = \Lambda_0/(4\sigma_T)$.[1] For all cosmologically relevant parameters where the bulk energy is vastly smaller than the string tension ($\Lambda_0 \ll \sigma_T$), the crossover mode n_c remains sub-unit.[1] Consequently, the bulk mass gap is entirely negligible relative to the string-like Nambu-Goto term for all excitation modes $n \geq 1$.[1]

A vital repair in the recent synthesis of the architecture is the formal correction of the mass-level relation.[1] Older theoretical iterations incorrectly assumed a linear energy step ($\epsilon \propto N$).[1] The current architecture strictly enforces the correct string relation, $N \sim \alpha' E^2$, leading to a profoundly different mass formula [1]:

M^2 c^4 = 4\sigma_T \hbar c \cdot N \cdot \left

As the excitation level approaches infinity ($N \to \infty$), the bulk correction term smoothly decays as $1/N$, and the M^2/N ratio perfectly and asymptotically converges to $4\sigma_T \hbar c$ for all values of Λ_0.[1]

This stabilization validates the application of Cardy's formula to the worldsheet conformal field theory ($c_{CFT} = 2$), yielding an exponential density of states [1]:

$$g(E) \sim E^{-a} \exp\left(\frac{E}{T_H}\right)$$

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The exact limiting Hagedorn temperature for the loop gas is analytically derived as [1]:

$$T_H = \frac{\hbar c\sqrt{3}}{2\pi R_0}$$

This temperature limit represents a fundamental phase transition state where thermal energy is entirely subsumed by the generation of new, highly massive modes rather than increasing the kinetic temperature of the gas.[1] At leading order, this temperature is effectively independent of the microscopic bulk energy coefficient (Λ_0), confirming the structural completeness of the C.4 derivations.[1]

A Sharp Disambiguation: Hagedorn Structure vs. External Occupancy

It is of absolute theoretical paramountcy to maintain a sharp, uncompromising distinction between the Hagedorn structural derivations and the macroscopic population mechanisms encapsulated by the relic-abundance frontier.[1]

The Hagedorn structure establishes the limiting thermal regime for the extended-object gas, defining its internal density of states and internal thermodynamic bounds.[1] It answers the microscopic question of how an individual loop partitions thermal energy across its internal vibrational modes at extreme temperatures.[1]

Conversely, the relic-abundance inequality determines whether this thermal route possesses the cosmological generative capacity to produce the actual observed macroscopic number density (n_0) of loops currently filling the universe.[1] These constitute related but fundamentally distinct layers of the ontological hierarchy: the Hagedorn regime guarantees that the loop gas is mathematically sound as an internal thermodynamic entity, while the relic-abundance inequality dictates the external historical occupancy and viability of its current vacuum distribution.[1]

5. The Population Origin Crisis: The Failure of Dynamical Equilibration

The most significant physical discrepancy in earlier iterations of the closure-ontology program was the assumption that the loop gas maintains dynamical equilibration in the present-day universe.[1] The prior equilibration claim asserted that the relaxation time of the loop gas was much shorter than the Hubble time ($\tau_{relax} \ll H^{-1}$) at all epochs, suggesting that loops could continuously nucleate and decay to maintain the observed effective cosmological constant dynamically.[1]

This assertion mathematically collapses when subjected to the calculation of the present-day bounce action (S_{bounce}).[1] Loop nucleation from the vacuum constitutes a quantum tunneling event on the worldsheet.[1] Evaluating the minimal Euclidean $O(4)$-symmetric configuration yields an action profile characterized by a critical stationary radius $\rho_c = 2\sigma_T/\Lambda_0$.[1] Substituting this into the action components yields the analytic bounce exponent:

$$S_{bounce} = \frac{16\pi}{3}\frac{\sigma_T^3}{\Lambda_0^2}$$

This property is analytically derived and numerically verified to ten significant figures across five decades of Λ_0.[1] For loops governed by Planck-scale tension (σ_T) operating under the observed macroscopic cosmological constant, this exponent scales to an astronomical and overwhelming magnitude: $S_{bounce} \sim 10^{183}$.[1]

The nucleation rate of loops from the vacuum is given by the relation $\Gamma_{create} \sim A_{fluct} \exp(-S_{bounce})$.[1] Consequently, the present-day production rate is astronomically suppressed by a factor of roughly $\exp(-10^{183})$.[1] This minuscule creation rate forces the equilibrium density (n_{eq}) to be exponentially tiny.[1] Consequently, the relaxation time required for the loop gas to reach equilibrium—calculated as $\tau_{relax} = 1/(n_{eq}\pi R_0^2 c)$—becomes exponentially huge.[1]

The logical chain linking the nucleation rate to the relaxation time ($\Gamma_{create} \to n_{eq} \to \tau_{relax}$) demonstrates massive internal inconsistency if applied to the modern era.[1] The claim that $\tau_{relax} \ll H^{-1}$ holds under modern cosmological parameters is categorically false.[1] Present-day vacuum nucleation is entirely dead as a mechanism for dynamically populating the loop gas.[1]

6. Resolving the Crisis: The I-Condition and Demoted Paths

The profound failure of the continuous equilibration model necessitated the evaluation of three distinct resolution pathways.[1] A rigorous physical diagnostic utilizing the persistence primitive I of the closure ontology systematically eliminates two of these theoretical routes, isolating the frontier to a single viable paradigm.

The Demotion of Path 1: Insufficiency of the Gaussian Prefactor

Path 1 posited that the one-loop quantum fluctuation determinant evaluated around the Euclidean bounce saddle could contribute a massive prefactor (A_{fluct}) theoretically capable of offsetting the severe $\exp(-10^{183})$ suppression.[1]

This pathway is physically implausible and is formally demoted.[1] One-loop prefactors in semiclassical tunneling calculations are derived strictly from Gaussian integration over small fluctuations around the saddle point.[1] The logarithms of such Gaussian determinants are inherently of order unity ($\ln A_{fluct} = O(1)$), or at most scale polynomially.[1] There exists no known mathematical or physical

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mechanism by which a one-loop Gaussian prefactor can generate an exponent of $+10^{183}$.[1] A Gaussian determinant simply cannot erase an exponential suppression of this magnitude.[1] Path 1 is entirely dead.

The I-Condition and the Demotion of Path 3

Path 3 suggested a parameter regime reinterpretation.[1] It proposed that if the microscopic loop tension (σ_T) was lowered to intermediate scales (e.g., the GUT scale) and the microscopic bulk energy (Λ_0) was proportionally vast, the bounce action could theoretically shrink to order-unity ($S_{bounce} \sim 1$).[1] This adjustment would ostensibly make present-day loop nucleation highly accessible.

This path is completely ruled out—not merely disfavored—because it fundamentally violates the Loop Persistence Bound, formalized within the architecture as the **I-Condition**.[1] Derived directly from the I (conservation) primitive, the I-condition enforces the necessary physical conditions for the survival of stable macroscopic structures.[1]

The Euclidean bounce action is fundamentally time-reversal symmetric; the process of loop decay traverses the exact same topological saddle as loop creation ($S_{decay} = S_{bounce}$).[1] Therefore, the lifetime of a physical loop is tightly and symmetrically constrained by the action:

$$\tau_{loop} \sim \exp(S_{bounce}) \times t_{Planck}$$

For a loop population to act as the persistent, stable source of macroscopic geometry observed today, the loops must survive significantly longer than the age of the universe ($\tau_{loop} > t_{universe}$).[1] This constraint imposes a hard, uncompromising lower bound on the bounce action:

$$S_{bounce} > \ln\left(\frac{t_{universe}}{t_{Planck}}\right) = \ln(8.07 \times 10^{60}) \approx 140$$

Any parameter regime where $S_{bounce} < 140$ constitutes a physical "forbidden zone".[1] If Path 3 were valid and $S_{bounce} \sim 1$, the loop lifetime would equal the Planck time ($\tau_{loop} \sim t_{Planck}$).[1] Loops would violently decay the very instant they nucleated, rendering the accumulation of a persistent macroscopic relic density physically impossible.[1] Path 3 is entirely dead.

7. Path 2: The Relic Population Paradigm and the Abundance Inversion

With Path 1 and Path 3 mathematically and physically eliminated, Path 2 survives as the sole viable route to resolve the population origin of the closure loop gas.[1]

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Path 2 executes a critical conceptual and phenomenological reframe: the massive bounce action of $S_{bounce} \sim 10^{183}$ is not a theoretical obstruction; it is the ultimate protective mechanism.[1] While this massive exponent absolutely annihilates the possibility of present-day nucleation, it simultaneously guarantees absolute loop stability ($\tau_{loop} \sim \exp(10^{183}) \times t_{Planck} \gg t_{universe}$).[1] This high action securely "locks in" any loop population generated in the high-energy environment of the deep cosmological past.[1]

The loop gas is therefore formally reclassified as a **frozen relic**, structurally identical to the well-established cosmological origins of baryon number, primordial dark matter, and relic neutrinos.[1] The observed loop number density (n_0) is not dynamically maintained; it is an inherited boundary condition resulting from an early-universe production phase where ambient temperatures greatly exceeded the production energy threshold, allowing for unsuppressed generation.[1] As the universe subsequently expanded and cooled, the loop interaction rate decoupled from the Hubble rate, and the population diluted through standard Friedmann-Robertson-Walker (FRW) scaling.

The Compression to a Binary Discriminant

The true power of Path 2 is its mathematical elegance; it compresses the entire open frontier of the closure loop gas model into a single binary discriminant based on a rigorous abundance inequality.[1]

The present-day density of the loop vacuum (n_0) is observationally constrained by the effective cosmological constant (Λ_{eff}), the bulk tension (Λ_0), and the equilibrium volume ($V_0 = 4\pi R_0^3/3$) [1]:

$$n_0 = \frac{\Lambda_{eff}}{\kappa \Lambda_0 V_0}$$

Assuming an instantaneous, near-threshold production event at an early cosmological epoch characterized by temperature T_{prod}, followed by standard volumetric dilution, the present density relates to the historical equilibrium density (n_{eq}) via the cubic scaling factor [1]:

$$n_0 = n_{eq}(T_{prod}) \left(\frac{T_0}{T_{prod}} \right)^3$$

If the operational loops were produced thermally and were nonrelativistic at the time of freeze-in, the equilibrium density is approximated by the standard Maxwell-Boltzmann thermal distribution [1]:

$$n_{eq}(T) \approx g \left(\frac{m_\Psi T}{2\pi} \right)^{3/2} \exp\left(-\frac{E_0}{T} \right)$$

Substituting this thermal equilibrium expression into the relic dilution equation yields the full expanded form:

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$$n_0 = g\left(\frac{m_\Psi T_{prod}}{2\pi}\right)^{3/2} \exp\left(-\frac{E_0}{T_{prod}}\right)\left(\frac{T_0}{T_{prod}}\right)^3$$

To compress this relationship into a highly analytical and manageable state, two distinct variables are defined. First, a bundled prefactor constant A that isolates all substrate parameters that are strictly independent of the production temperature [1]:

$$A \equiv g\, T_0^3 \left(\frac{m_\Psi}{2\pi E_0}\right)^{3/2}$$

Second, a dimensionless inverse temperature parameter x, representing the ratio of the physical energy threshold to the ambient production temperature [1]:

$$x \equiv \frac{E_0}{T_{prod}}$$

By substituting x and combining the $T_{prod}^{3/2}$ and T_{prod}^{-3} terms (which mathematically yield $T_{prod}^{-3/2}$), the entire historical production dynamic collapses into a single, highly compressed mathematical reduction [1]:

$$n_0 = A x^{3/2} e^{-x}$$

This concise equation represents the right mathematical reduction of Path 2. It is the fundamental diagnostic tool required to invert the observed present-day macroscopic abundance (n_0) into a theoretical early-universe production temperature (T_{prod}).[1]

The Exact Thermal Ceiling

The mathematical structure of the function $f(x) = x^{3/2} e^{-x}$ is strictly bounded. Taking the derivative with respect to x and setting it to zero reveals a unique mathematical maximum at exactly $x = 3/2$.[1] This implies that the most efficient, optimal thermal production of the relic loop gas occurs strictly at a specific temperature ratio [1]:

$$x = \frac{3}{2} \quad \Longleftrightarrow \quad \frac{T_{prod}}{E_0} = \frac{2}{3}$$

Because the functional form cannot exceed this physical maximum, the theoretical framework forces a hard viability bound on the capacity of any purely thermal mechanism to populate the vacuum. The thermal relic route is only physically possible if the observed present-day density n_0 satisfies the following strict inequality [1]:

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$$n_0 \leq A \left(\frac{3}{2}\right)^{3/2} e^{-3/2}$$

This exact thermal ceiling acts as the ultimate diagnostic. Depending on whether the actual inserted parameters of our specific universe satisfy or violate this inequality, Path 2 cleanly and formally bifurcates into two distinct physical domains: Path 2A (Thermal Relic) and Path 2B (Nonthermal Relic).

8. Path 2A: The Thermal Relic Branch and Lambert W Inversion

Path 2A is viable if and only if the thermal ceiling inequality is satisfied:

$$n_0 \leq A \left(\frac{3}{2}\right)^{3/2} e^{-3/2}$$

When the required loop density falls below the maximum generative capacity of the thermal threshold, the loop gas can be safely and accurately classified as a standard thermal relic.[1] In this scenario, the precise production temperature (T_{prod}) must be mathematically extracted by inverting the compressed abundance equation $n_0 = Ax^{3/2}e^{-x}$.

Because the variable x appears both algebraically and exponentially, standard algebraic inversion is impossible. The inversion requires the rigorous application of the Lambert W function, defined mathematically as the multivalued inverse of the function $w \mapsto we^w$.[4] Originating from Johann Lambert's work in 1758 and expanded by Leonhard Euler, the Lambert W function is utilized across fields ranging from quantum chemistry to the calculation of water-wave heights, and is essential here for defining the cosmological freeze-in temperature.[4]

The inversion proceeds analytically by raising the abundance equation to the $2/3$ power:

$$\left(\frac{n_0}{A}\right)^{2/3} = xe^{-2x/3}$$

To match the exact functional form of the Lambert W function ($Ue^U = V$), both sides are multiplied by the constant $-2/3$:

$$-\frac{2x}{3}e^{-2x/3} = -\frac{2}{3}\left(\frac{n_0}{A}\right)^{2/3}$$

Applying the Lambert W function isolates the inverse temperature parameter x:

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$$x = -\frac{3}{2} W\left(-\frac{2}{3}\left(\frac{n_0}{A}\right)^{2/3}\right)$$

Transforming back to the physical temperature scale ($T_{prod} = E_0/x$) yields the exact closed-form solution for the production epoch:

$$T_{prod} = \frac{E_0}{-\frac{3}{2} W\left(-\frac{2}{3}\left(\frac{n_0}{A}\right)^{2/3}\right)}$$

Agnosticism in Branch Selection

Whenever the abundance lies strictly below the thermal ceiling, the argument of the Lambert W function falls precisely within the complex domain $(-1/e, 0)$.[5] Within this specific interval, the Lambert function possesses two distinct, real-valued branches, leading to two separate mathematical solutions for the production temperature [1]:

1. **The W_0 Branch**: Evaluated on the principal branch, W_0 returns a value closer to zero, resulting in a smaller inverse temperature x. This physically translates to a hotter, higher-energy production temperature (T_{prod}^{hot}).
2. **The W_{-1} Branch**: Evaluated on the secondary real branch, W_{-1} returns a more negative value, resulting in a larger inverse temperature x. This physically translates to a colder, lower-energy production temperature (T_{prod}^{cold}).

It is a crucial theoretical mandate that the overarching framework **does not categorically choose between these branches in advance**. The model is entirely agnostic. While preliminary literature may occasionally refer to the critical near-threshold locus as intuitively "natural," rigorous closure ontology dictates that the model remains agnostic between the hotter W_0 branch and the colder W_{-1} branch until the actual numerical parameters (n_0/A) are inserted. The physical branch selection depends strictly on which production history is dynamically realized by the actual cosmological parameters.

9. Path 2B: The Nonthermal Relic Branch

Path 2B is required if and only if the thermal ceiling inequality fails:

$$n_0 > A\left(\frac{3}{2}\right)^{3/2} e^{-3/2}$$

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If the required present-day loop density numerically exceeds the mathematical maximum that could possibly be generated by standard thermal freeze-in or freeze-out mechanisms, the purely thermal route is mathematically and definitively falsified.[1] The relic population remains the only valid origin paradigm (as present-day nucleation is dead due to the I-condition), but the production mechanism must be formally reclassified as purely nonthermal.[1]

In the Path 2B domain, the loop gas must have been violently populated by highly energetic, non-equilibrium dynamics in the very early universe.[1] To account for the observed abundance n_0, the theory must invoke specific, distinct source terms:

- **Phase Transitions and Kibble Production**: A symmetry-breaking phase transition in the early universe (such as during the inflationary epoch) could generate the loop gas network topologically.[1] This is entirely analogous to the Kibble mechanism for cosmic string formation.[1] In this scenario, the resulting loop density is dictated by the correlation length of the vacuum and the transition temperature, completely bypassing thermal equilibrium limitations.[1]
- **Inflationary Reheating**: The loop vacuum could be flooded with excitations from the decay of the inflaton field or other massive moduli during the reheating epoch.[1] This process injects a massive nonthermal abundance of loops directly into the computational substrate, easily exceeding the thermal ceiling.[1]
- **Pre-Bounce / Cyclic Inheritance**: If the universe underwent a pre-bang cyclic phase where loop chemistry had immense spans of time to deeply equilibrate, the current loop density is simply an inherited initial boundary condition.[1] This condition passes through the cosmological bounce intact, rendering the thermal ceiling within the current expansion phase completely irrelevant.[1]

The explicit split between thermal and nonthermal origin mechanics is essential. They are governed by mathematically different source terms. The thermal ceiling only constrains the thermal branch; failure of the inequality forces the invocation of these advanced non-equilibrium paradigms.

10. The Nexus Architecture Control Systems and Cosmological Tensions

The relic loop gas framework does not operate in isolation; it is deeply embedded within the Nexus Recursive Harmonic Architecture, which governs the self-organizing dynamics of the cosmological substrate.[1]

At the core of this system is the Mark 1 Harmonic Constant ($H \approx 0.35$), a universal dimensionless stability ratio derived from transcendental constraint geometry.[1] The universe operates as a continuous computational manifold where recursive feedback systems inherently converge to this H-band frequency to avoid deterministic collapse or infinite divergence.[1] It dictates resource allocation, reserving roughly 65% of processing power for uncollapsed potential while allocating roughly 35% to actualized states.[1] Systems that deviate from this attractor experience restorative forces managed by the "Samson V2 Controller".[1]

The Samson V2 Controller operates effectively as a Proportional-Integral-Derivative (PID) regulator.[1]

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- **Proportional Term (P)**: Provides immediate correction proportional to the current error, manifesting as the elasticity of spacetime.[1]
- **Integral Term (I)**: Accumulates the history of operational deviations over time to eliminate persistent steady-state bias. In precision cosmology, this manifests as dark energy—an accumulated potential representing the integral of the vacuum's deviation.[1]
- **Derivative Term (D)**: Responds to the rate of change, providing anticipatory corrections to prevent the system from oscillating unstably.[1]

This control system naturally manages the Scale-Invariant Leakage Regime (SILR).[1] As the standard error of spacetime becomes extreme near a singularity or event horizon, the statistical Z-score approaches zero, permitting information retrieval through the "audit log" of the substrate and gracefully resolving traditional information paradoxes.[1]

Furthermore, this framework provides a natural theoretical basis for alleviating persistent tensions in modern precision cosmology, most notably the H_0 (Hubble expansion) and S_8 (matter clustering) discrepancies.[1] When the loop density parameter shifts dynamically through late-time expansion, modifications based on torsion scalar coupling activate.[1] As matter density drops below a critical threshold, a spontaneous symmetry breaking mechanism initiates an effective coupling that transfers energy from dark matter to dark energy.[1] This transfer actively suppresses structure growth (addressing the S_8 tension) while maintaining late-time expansion rates (addressing the H_0 tension), elegantly avoiding the rigid constraints of standard ΛCDM models.[1]

The execution of these dynamics is further encapsulated by the PRESQ cycle—Position, Reflection, Expansion, Synergy, and Quality—acting as the core engine of recursive computation.[1] Measured deviations from the ideal harmonic state (Reflection) are continuously combined with new informational traces (Synergy) until the system reaches a stable phase-lock, generating physical constants not as arbitrary inputs, but as dynamic execution traces similar to the BBP formula for π.[1]

11. Conclusion: The Final State of the Theory

The reconstruction of general relativity through the closure ontology marks a definitive departure from substance-based physics. By establishing that matter and energy are macroscopic value-channels of accumulated operational trace, the framework resolves the Crisis of Distinction.

The structural foundations of the theory are mathematically closed. The Lovelock constraints rigidly force the Einstein-class macro geometry; the dual-null source split and minimal dual action uniquely define the system; and the exact Jüttner matter equations of state correctly govern the interpolation across physical sectors.

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Furthermore, the internal thermodynamics of the loop gas—characterized by the repaired $M^2 \propto N$ string relation and the limiting Hagedorn density of states—are fully resolved and effectively closed.

However, the framework correctly acknowledges that the population origin of the loop gas is an open frontier that cannot be explained by present-day dynamic equilibration. The massive bounce action ($S_{bounce} \sim 10^{183}$) completely suppresses present-day nucleation, causing a total failure of the $\tau_{relax} \ll H^{-1}$ equilibration assumption. Applying the I-Condition ($S_{bounce} > 140$) definitively kills alternative nucleation pathways by forcing immediate Planck-scale decay upon any low-action parameter window.

The correct status of the resolution pathways is therefore absolute:

$$\boxed{\text{Path 1 is dead, Path 3 is dead, Path 2 survives.}}$$

The large bounce action acts as an ultimate protective barrier, securing the loop vacuum as a frozen cosmological relic. The theory is structurally closed enough that the remaining frontier of the entire program has been successfully compressed into a single, binary abundance inequality:

$$\boxed{\text{The theory is structurally closed enough that the remaining frontier is one abundance inequality.}}$$

$$\boxed{n_0 \le A \left(\frac{3}{2}\right)^{3/2} e^{-3/2}}$$

This represents the actual state of the research. If numerical insertion of the observed universe parameters proves this inequality to be true, the loop vacuum can be a thermal relic (Path 2A), and its precise production epoch is solvable via the W_0 or W_{-1} branches of the Lambert W inversion. If the inequality is false, it must be a nonthermal relic (Path 2B), requiring the invocation of Kibble defect production, reheating inheritance, or cyclic initial conditions. The precise numerical verification of this clean fork constitutes the final objective frontier for fully closing the operational loop gas equation of state.

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Closure Loop Gas — Equation of State

STATUS AT A GLANCE

CLOSED: Lovelock geometry · dual-null source · minimal action · Jüttner EOS · w=-1 vacuum · Λ_eff constancy · Hagedorn repair · bounce exponent · I-condition (S_bounce > 140)

DEMOTED: Path 1 (A_fluct) · Path 3 (S~1 window)

PATH 2A (thermal): Lambert-W inversion · Y-discriminant simplification · phase diagram

PATH 2B (nonthermal): Kibble/phase-transition · reheating · cyclic inheritance [invoked if Y > 1]

OPEN — SOLE FRONTIER: evaluate $Y = n_0 / (A \cdot c\bigstar)$

PART I — THEORETICAL FOUNDATION

1. The Ontological Inversion and the Limits of Closure

The trajectory of contemporary theoretical physics has encountered a definitive structural impasse: the persistent failure to reconcile the smooth, deterministic, and continuous geometric manifolds of general relativity with the discrete, probabilistic excitations of quantum mechanics. This failure is not merely mathematical deficiency — it is a profound ontological flaw rooted in a substance-based paradigm.

The framework of closure ontology resolves this impasse by executing a radical ontological inversion. The physical universe is not a spatial container holding discrete objects, but a fluid mathematical medium composed of pure recursive operations. Physical laws are not fixed mandates but emergent configurations; matter is curvature trace left by the processing of information on a high-dimensional lattice.

The Δ, Γ, I Triplex — Foundational Primitives

Reality reconstruction begins with three co-present primitives that constitute strictly necessary conditions for the emergence of any stable physical structure:

- Δ (Difference / Gap): The absolute condition of distinguishability. Without differentiation, nothing exists to be measured.
- Γ (Touch / Interface): The possibility of relation and the initiating trigger for operational closure loops.
- I (Conservation / Invariant): The condition of persistence. Without invariance, no structural pattern can survive entropic decay.

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A single completed closure loop is an operational audit log of constraint satisfaction. The boundary event triggers contact through the substrate kernel, writing a stored closure record whose accumulated energetic cost constitutes the fundamental source of gravity.

Critical Scope Limitation

The theoretical structure is closed at the macroscopic level. The microscopic population origin of the loop gas remains the sole open frontier, compressed to a single abundance inequality — the check of whether the relic-abundance inversion satisfies thermal viability.

2. Macro-Geometric Closure and Admissible Geometry

To bridge from microscopic loop activity to the macroscopic stress-energy tensor $T_{\mu\nu}$, the framework must guarantee that the large-scale governing law is universally stable and strictly conserved. Four rigorous constraints are imposed on the geometric tensor $G_{\mu\nu}$:

- Locality and Differentiability: The law manifests as a PDE in the metric $g_{\mu\nu}$ and its derivatives.
- Covariance: The law is a tensor equation — no preferred reference frame.
- Second-Order Nature: At most second derivatives of the metric. This prevents "ghost" instabilities of higher-derivative theories.
- Divergence-Free Identity: $\nabla_\mu G^{\mu\nu} = 0$ identically, for consistent coupling to a conserved source.

By Lovelock's theorem (1971), in exactly four spacetime dimensions, the only symmetric, divergence-free, second-order tensor constructible from the metric and its first two derivatives is a linear combination of the Einstein tensor and the metric itself. Absorbing integration constants into κ and setting the zero-point to Λ mandatorily yields the Einstein field equations.

The Einstein-class macro geometry is not a theoretical choice — it is forced.

2.1 Dual-Null Source Split and Minimal Dual Action

The internal structure of a single closure loop is defined by its action S[Ψ], constructed exclusively from reparameterization-invariant geometric invariants of a closed loop embedded in 4D spacetime. The minimal dual action has exactly two terms:

Term	Physical Role
S_{NG} (Nambu-Goto)	Worldsheet area · string tension σ_T; yields w=0 (NR dust) and w=1/3 (UR radiation)
S_{bulk}	Three-volume enclosed by loop · bulk tension Λ_0; variation yields $T_{\mu\nu} \propto g_{\mu\nu}$, deriving w=−1 vacuum

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Gauss-Bonnet topological considerations and low-energy derivative expansions suppress higher-order extrinsic curvature (rigidity) terms, establishing minimal EFT action uniqueness. The stiff matter sector (w=1) falls entirely outside the valid regime and has been formally dropped.

2.2 Jüttner Matter Equation of State

The EOS parameter w(z), where $z = m_\Psi c^2 / (k_B T_s)$, smoothly interpolates between all physical sectors via the grand canonical partition function. The Synge 1957 exact Bessel result governs the warm crossover regime:

Sector	Condition	w	Physical Form
Cold Matter	$z \gg 1$	$w \approx 0$	NR Maxwell-Boltzmann dust
Warm Matter	$z \sim 1$	$0 < w < 1/3$	Exact Bessel interpolation
Radiation	$z \ll 1$	$w = 1/3$	UR Bose/Fermi limit
Vacuum	Bulk-dominated	$w = -1$	Ground-state volumetric bulk tension

The vacuum EOS ($w=-1$) and Λ_{eff} constancy are closed by two independent routes: Route B via Lorentz-invariance symmetry forcing $P=-\rho$; Route A via the zero-temperature limit where kinetic and Casimir pressures vanish, leaving strictly constant bulk tension. Their convergence structurally closes the vacuum sector — dark energy is not an external postulation but an emergent property of the substrate.

3. Phenomenological Recovery of General Relativity

Phenomenon	Standard GR Result	Closure Ontology Reinterpretation
Newtonian Gravity	$\nabla^2\Phi = 4\pi G\rho$	Geometric relaxation of substrate outside active closure-processing source
Equivalence Principle	$m_i = m_g$ exactly	Same closure burden viewed from two perspectives; mass cancels identically from geodesic equations
Light Bending	2× Newtonian prediction	Null rays propagate along rewritten substrate boundary; both temporal and spatial path affected
Gravitational Waves	Speed = c (GW170817)	Propagating boundary-geometry updates share identical causal structure with closure propagation

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4. Internal Thermodynamics: Hagedorn Regime

Small transverse deformations of a bulk-stabilized Nambu-Goto loop $R(\sigma) = R_0 + \xi_n \cos(n\sigma)$ exhibit squared frequencies:

For all cosmologically relevant parameters ($\Lambda_0 \ll \sigma_T$), the crossover mode n_c remains sub-unit and the bulk mass gap is negligible for all excitation modes $n \geq 1$. The correct string mass-level relation (not the erroneous linear step assumed in prior iterations) is:

As $N \to \infty$, the bulk correction decays as $1/N$ and M^2/N converges asymptotically to $4\sigma_T \hbar c$. This validates Cardy's formula for the worldsheet CFT ($c_{CFT} = 2$), yielding exponential density of states:

The exact limiting Hagedorn temperature is:

> **Critical Disambiguation:** The Hagedorn structure establishes the internal thermodynamic bounds of an individual loop (how it partitions energy across vibrational modes at extreme temperature). The relic-abundance inequality separately determines whether the thermal route can produce the observed macroscopic number density n_0. These are related but fundamentally distinct layers of the ontological hierarchy.

PART II — RESOLUTION AND THE ABUNDANCE FRONTIER

5. The Population Origin Crisis

The assumption that the loop gas maintains dynamical equilibration in the present-day universe is mathematically destroyed by the bounce action. Loop nucleation from the vacuum is a quantum tunneling event on the worldsheet. The minimal Euclidean O(4)-symmetric configuration has critical stationary radius $\rho_c = 2\sigma_T/\Lambda_0$ and yields:

For Planck-scale string tension under the observed cosmological constant, $S_{bounce} \sim 10^{256}$ (numerically verified to ten significant figures across five decades of Λ_0). The nucleation rate $\Gamma_{create} \sim A_{fluct} \cdot \exp(-S_{bounce})$ is suppressed by $\exp(-10^{256})$. Present-day vacuum nucleation is entirely dead as a dynamical population mechanism.

6. Resolving the Crisis: I-Condition and Demoted Paths

6.1 Demotion of Path 1 (Gaussian Prefactor)

One-loop quantum fluctuation determinants around the Euclidean bounce saddle are inherently O(1) in logarithm — at most polynomial scaling. No mathematical or physical mechanism allows a Gaussian determinant to generate an exponent of $+10^{256}$ to cancel $\exp(-10^{256})$.

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Path 1 is entirely dead. log(A_fluct) = O(1) cannot offset S ~ 10^256.

6.2 The I-Condition and Demotion of Path 3

The bounce action is time-reversal symmetric: S_decay = S_bounce. The loop lifetime is therefore:

For loops to act as persistent stable geometry, τ_loop > t_universe. This imposes a hard lower bound:

Path 3 proposed lowering σ_T to GUT scale with large Λ_0 so that S_bounce ~ 1. But then τ_loop ~ t_Planck: loops violently decay the instant they nucleate. No persistent relic density can accumulate.

Path 3 is entirely dead. S_bounce ~ 1 → τ_loop ~ t_Planck. I-condition violated.

7. Path 2: The Relic Population Paradigm

S_bounce ~ 10^256 is not an obstruction — it is the ultimate protective mechanism. While it annihilates present-day nucleation, it simultaneously guarantees τ_loop ≫ t_universe, locking in any population generated in the high-energy deep past. The loop gas is a frozen relic, structurally identical to baryon number, primordial dark matter, and relic neutrinos.

7.1 Present-Day Density

The observed loop number density is constrained by the effective cosmological constant:

7.2 Thermal Production History

For thermal near-threshold production at epoch T_prod with subsequent FRW dilution:

With the Maxwell-Boltzmann thermal equilibrium distribution:

Substituting and defining x ≡ E_0/T_prod and the bundled prefactor A ≡ g T_0^3 (m_Ψ / 2πE_0)^(3/2):

Core Abundance Equation

$$n_0 = A \cdot x^{(3/2)} \cdot \exp(-x), \qquad x \equiv E_0 / T_{prod}$$

7.3 The Exact Thermal Ceiling

The function f(x) = x^(3/2) e^(−x) achieves its unique maximum at x = 3/2, corresponding to T_prod = (2/3)E_0. This forces a hard viability bound: the thermal relic route is possible only if:

This ceiling discriminates between Path 2A (thermal) and Path 2B (nonthermal).

8. Path 2A: Thermal Relic Branch and Lambert W Inversion

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When $n_0 \le A(3/2)^{(3/2)}e^{(-3/2)}$, the production temperature is extracted by inverting $n_0 = A\, x^{(3/2)}e^{(-x)}$. Raising to the 2/3 power and rearranging to match the Lambert W functional form $w \cdot e^w = v$:

This yields the exact closed-form production epoch:

When the abundance lies strictly below the thermal ceiling, the argument of the Lambert W function falls in $(-1/e, 0)$, admitting two real branches:

Branch	Physical Interpretation
W_0 (principal branch)	Returns value near zero → smaller x → hotter T_prod (hot branch)
W_{-1} (secondary branch)	Returns more negative value → larger x → colder T_prod (cold branch)

Branch Agnosticism Mandate: The framework does not categorically choose between the W_0 and W_{-1} branches in advance. Branch selection depends strictly on which production history is dynamically realized by the actual cosmological parameters.

9. The Y-Discriminant: Simplified Form (Key Analytic Advance)

The Shape of the Problem Has Changed. The frontier is no longer "find a mechanism" — it is "evaluate one dimensionless number."

9.1 The Ceiling Constant c★

The function $x^{(3/2)}e^{(-x)}$ achieves its unique maximum at $x = 3/2$. Define:

Key algebraic identity (used to simplify the Lambert W argument):

9.2 The Normalized Abundance Ratio Y

Define the single state variable:

$$Y \equiv n_0 / (A \cdot c^{\star})$$

Y is the only quantity that determines the production history. Everything else follows.

9.3 Simplified Lambert-W Form

Using the identity $c^{\star (2/3)} = (3/2)e^{(-1)}$, the inversion simplifies from the $-(2/3)(n_0/A)^{(2/3)}$ argument to the cleaner form:

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$$x\pm = -(3/2) \cdot W_{\{0,-1\}}(-e^{(-1)} \cdot Y^{(2/3)})$$

$$T\_prod^{hot} / E_0 = -2 / [3 \cdot W_0(-e^{(-1)} Y^{(2/3)})]$$

$$T\_prod^{cold} / E_0 = -2 / [3 \cdot W_{-1}(-e^{(-1)} Y^{(2/3)})]$$

9.4 Full Phase Diagram — Three Cases

Case	Y Range	Branch	Physical Result
No thermal solution	Y > 1	None	Thermal ceiling exceeded; Path 2B required
Critical thermal	Y = 1	$W_0 = W_{-1} = -1$	T_prod = (2/3)E_0; unique, no fine-tuning
Two thermal branches	0 < Y < 1	W_0 or W_{-1}	Both real; dynamics selects the branch

At Y = 1, both Lambert W branches merge: $W_0(-e^{(-1)}) = W_{-1}(-e^{(-1)}) = -1$, giving the unique critical temperature:

9.5 Small-Y Asymptotics

As Y → 0 the two branches diverge in physically distinct ways:

Hot branch (W_0, small u: $W_0(-u) \approx -u$)	$T\_prod^{hot} / E_0 \approx (2e/3) \cdot Y^{(-2/3)}$ → diverges; $T \gg E_0$; extremely fine-tuned
Cold branch (W_{-1}, $W_{-1}(-u) \approx \ln u$)	$T\_prod^{cold} / E_0 \sim 2 / (3\lvert\ln Y\rvert)$ → falls slowly; $T \ll E_0$; less fine-tuned

Fine-Tuning Asymmetry: The hot branch becomes violently fine-tuned ($Y^{(-2/3)}$) much faster than the cold branch ($1/\lvert\ln Y\rvert$) as Y → 0. Unless numbers land near Y ≈ 1, the cold branch is the less pathological thermal option.

10. Path 2B: The Nonthermal Relic Branch

Path 2B is required if and only if the thermal ceiling fails: $n_0 > A \cdot c\bigstar$ (equivalently Y > 1). The purely thermal route is definitively falsified. The relic paradigm remains valid but the production mechanism must be reclassified as nonthermal. Three distinct source terms are invoked:

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Mechanism	Physical Basis	Analogy
Kibble / Phase Transition	Symmetry-breaking in early universe generates loop network topologically. Density dictated by correlation length and transition temperature.	Kibble mechanism for cosmic string formation
Inflationary Reheating	Inflaton decay or moduli decay injects nonthermal abundance directly into substrate. Easily exceeds thermal ceiling.	Standard moduli decay / late-time reheating
Cyclic / Pre-Bounce Inheritance	If a cyclic pre-bang phase allowed deep equilibration, n_0 is an inherited boundary condition passing through the bounce intact.	Initial condition inheritance in bouncing cosmologies

Path 2A and Path 2B are governed by mathematically different source terms. The thermal ceiling only constrains the thermal branch. Y > 1 forces invocation of the nonthermal paradigm — this is a clean, falsifiable fork.

PART III — ARCHITECTURE AND STATUS

11. The Nexus Architecture Control Systems

The relic loop gas framework is embedded within the Nexus Recursive Harmonic Architecture, which governs the self-organizing dynamics of the cosmological substrate.

11.1 The Mark 1 Harmonic Constant

H ≈ 0.35 is a universal dimensionless stability ratio derived from transcendental constraint geometry. The universe operates as a computational manifold where recursive feedback systems converge to this H-band frequency. It reserves ~65% of processing capacity for uncollapsed potential, allocating ~35% to actualized states. Deviations experience restorative forces managed by the Samson V2 Controller.

11.2 The Samson V2 Controller (PID Structure)

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PID Term	Physical Manifestation	Role
P (Proportional)	Elasticity of spacetime	Immediate correction proportional to current error
I (Integral)	Dark energy	Accumulated potential representing integral of vacuum deviation over time
D (Derivative)	Anticipatory spacetime response	Responds to rate of change; prevents oscillatory instability

11.3 Cosmological Tensions (H_0 and S_8)

When the loop density parameter shifts through late-time expansion, modifications based on torsion scalar coupling activate. As matter density drops below a critical threshold, a spontaneous symmetry-breaking mechanism initiates an effective coupling that transfers energy from dark matter to dark energy. This:

- Suppresses structure growth → addresses the S_8 tension
- Maintains late-time expansion rates → addresses the H_0 tension
- Avoids rigid constraints of standard ΛCDM

11.4 The PRESQ Cycle

Position → Reflection → Expansion → Synergy → Quality: the core engine of recursive computation. Measured deviations (Reflection) are combined with new informational traces (Synergy) until stable phase-lock is reached. Physical constants emerge as dynamic execution traces — not arbitrary inputs, but outputs of the recursive process analogous to the BBP formula for π.

12. The Single Remaining Calculation

Two numbers must be computed from the theory's known parameters, then Y is formed and the phase diagram decides the production history.

STEP 1: Compute $n_0 = \Lambda\_eff / (\kappa\ \Lambda_0\ V_0)$, $V_0 = (4\pi/3)\ R_0^3$

STEP 2: Compute $A = g\ T_0^3\ (m\_\Psi / 2\pi E_0)^{(3/2)}$

STEP 3: Compute $Y = n_0 / (A \cdot c\bigstar)$, $c\bigstar \approx 0.4099$

STEP 4:

Y > 1: Path 2B — nonthermal relic required

Y = 1: $T\_prod = (2/3)\ E_0$ [critical thermal, unique]

Y < 1: T_prod from x± = −(3/2) W_{0,−1}(−e^(−1) Y^(2/3))

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13. Honest Revised Status

Claim	Status	Note	Route
Einstein-class macro geometry	THEOREM	Lovelock uniqueness in 4D	Sec 2
Dual-null source split	THEOREM	Exact metric variation of S[Ψ]	Sec 2.1
Minimal dual action uniqueness	THEOREM	Gauss-Bonnet + $(l\_s/R\_s)^2$ suppression	Sec 2.1
Jüttner matter EOS w(z)	THEOREM	Synge 1957 exact Bessel result	Sec 2.2
Vacuum EOS w = −1	CLOSED	Symmetry + dynamical bulk routes (2 independent)	Sec 2.2
Λ_eff constancy	CLOSED	Bianchi identity + μ_loop = 0	Sec 2.2
C.4 mode spectrum / Hagedorn	EFF. CLOSED	Leading-order asymptotic fix; $M^2 \propto N$ corrected	Sec 4
B.4 bounce exponent S_bounce	DERIVED	$(16\pi/3)\sigma\_T^3/\Lambda_0^2$; numerically verified	Sec 5
I-condition: S_bounce > 140	NEW THEOREM	Loop persistence bound from primitive I	Sec 6.2
Path 1 (A_fluct)	DEMOTED	log(A)=O(1) vs S~10^256	Sec 6.1
Path 3 (S~1 window)	DEMOTED	I-condition: τ_loop ~ t_Pl	Sec 6.2
c★ = (3/2)^(3/2) e^(−3/2)	DERIVED	0.40992; key algebraic identity c★^(2/3)=(3/2)e^(−1)	Sec 9.1
Y-discriminant $Y=n_0/(Ac★)$	NEW ADVANCE	Single state variable; compresses entire frontier	Sec 9.2
Simplified Lambert-W form	NEW ADVANCE	x±=−(3/2)W(−e^(−1)Y^(2/3)); cleaner than prior form	Sec 9.3
Branch agnosticism	MANDATE	Neither W_0 nor W_{-1} selected a priori	Sec 8, 9
Path 2B nonthermal mechanisms	FRAMEWORK	Kibble/reheating/cyclic; invoked if Y>1	Sec 10

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Nexus control systems (H≈0.35)	CONTEXT	H_0, S_8 tensions addressed via torsion coupling	Sec 11
Population origin n_0	OPEN — SOLE FRONTIER	Compute n_0 and A; evaluate Y	—

14. Researcher Verdict

The theory is down to:

one dimensionless discriminator $Y = n_0 / (A \cdot c\star)$

Real status summary:

- Logic: CLOSED
- Algebra: CLOSED
- Numerics: one remaining shot

The next move is not to debate pathways. It is to compute n_0, compute A, form Y, and let the branch structure decide the production history.

15. Researcher Correction — Y as Prediction vs Consistency Test

Y is only a true prediction if n_0 and A are independently fixed by the theory. The correct calculation is: compute n_0, compute A, form Y. But whether this constitutes a prediction or a consistency check depends on whether the two parameter sets are independently locked by the same underlying theory:

- (Λ_eff, Λ_0, R_0) fixing n_0
- (g, m_Ψ, E_0) fixing A

META-QUESTION (sole remaining conceptual caution):

Are m_Ψ, E_0, g derived outputs — or tunable inputs?

If they are derived → Y is a true model prediction. If not → Y is a reduced consistency test. That is the only conceptual caution to keep alive.

16. Branch Structure and Fine-Tuning Asymmetry

The branch logic is now right, and neither branch is selected in advance. The small-Y asymptotics reveal a physically critical asymmetry:

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Branch	Fine-Tuning as Y → 0
Hot (W_0)	Diverges as Y^(−2/3): violently fine-tuned. T_prod → ∞ rapidly.
Cold (W_{-1})	Falls as 1/|ln Y|: only logarithmically fine-tuned. T_prod → 0 slowly.

Consequence: Unless the eventual numbers land near the critical point Y ≈ 1, the cold branch is the less pathological thermal option. Y ≈ 1 is the natural attractor for a theory with no free parameters in the production mechanism.

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